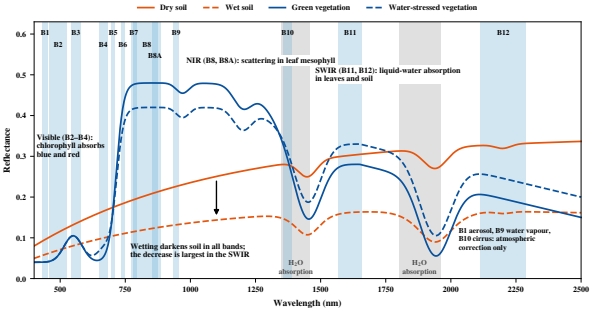
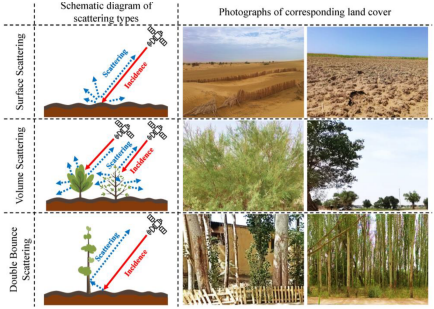
Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\text{SM}, \text{roughness}, \text{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	$\text{SM} = f_\theta(\text{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

Motivation Multimodal remote sensing



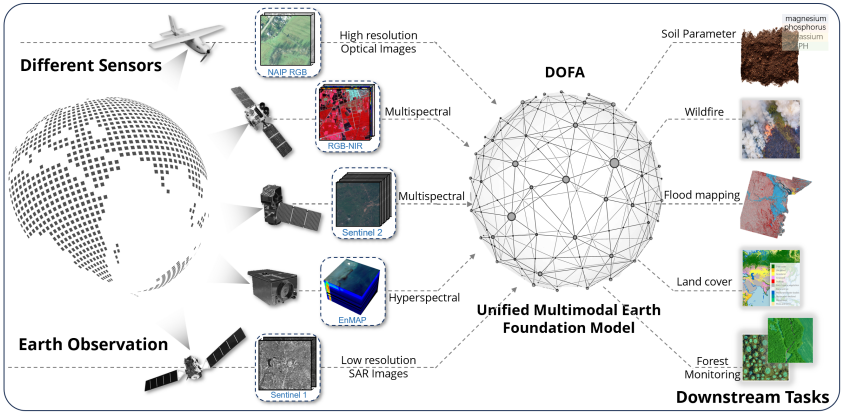
Optical: dry vs wet soil; green vs water-stressed vegetation



SAR scattering types (Nurmemet et al., 2025)

Soil and vegetation water change both optical reflectance and SAR backscatter

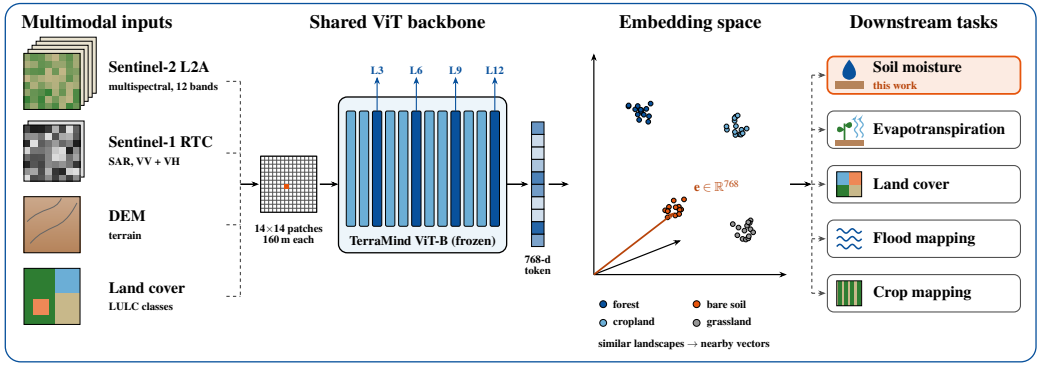
Motivation Geospatial foundation models (GFMs)



DOFA: one model for many sensors and many downstream tasks (Xiong et al., 2024, arXiv:2403.15356)

Once pre-trained on multimodal, unlabelled data, it works for many downstream tasks

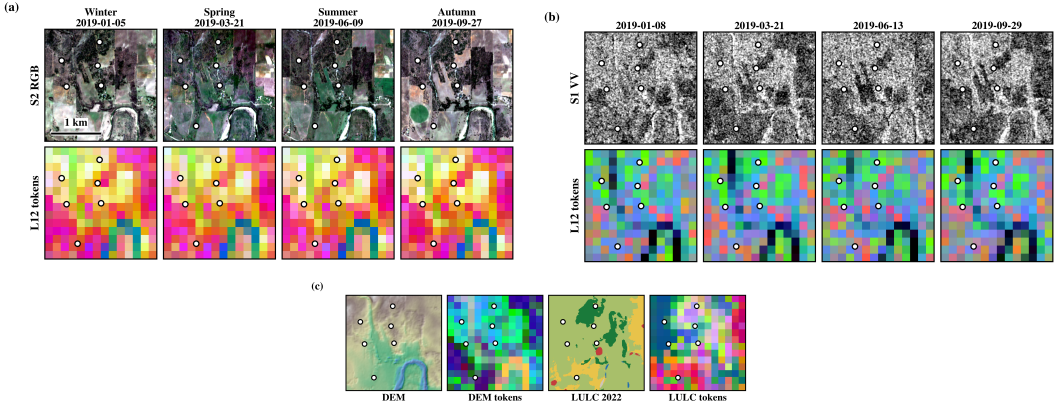
Motivation TerraMind embeddings for soil moisture



S2, S1, DEM and LULC → frozen TerraMind ViT-B → one 768-d embedding per 160 m patch → downstream tasks

The same embedding can serve many tasks; in this case, soil moisture

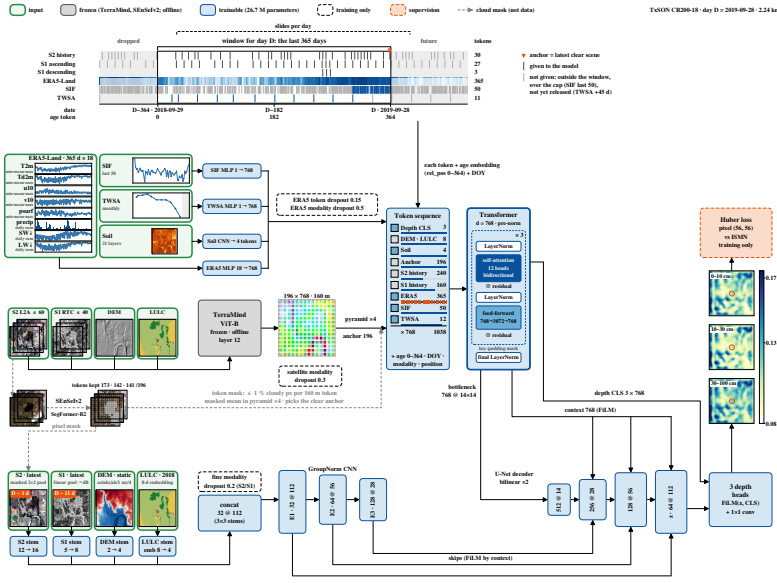
Motivation Example: TerraMind embeddings at one site (TxSON, 2019)



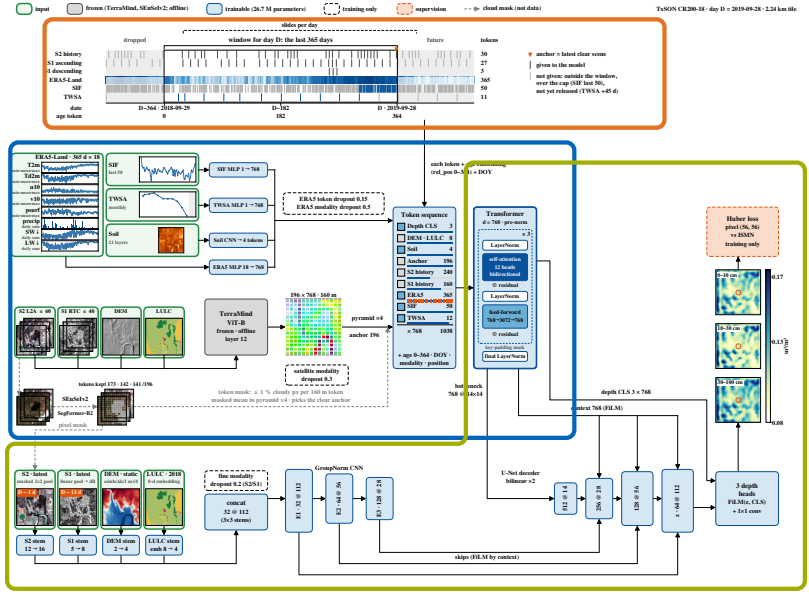
(a) S2 in four seasons, (b) S1 VV, (c) DEM and LULC, each with its TerraMind L12 tokens

Each 160 m patch → one 768-d vector; colours show its first 3 principal components

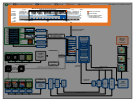
Method Model architecture

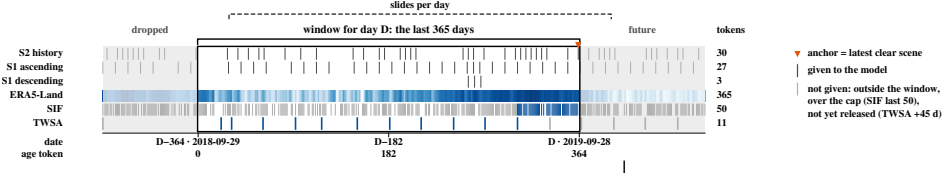


Method Model architecture

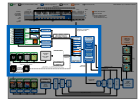


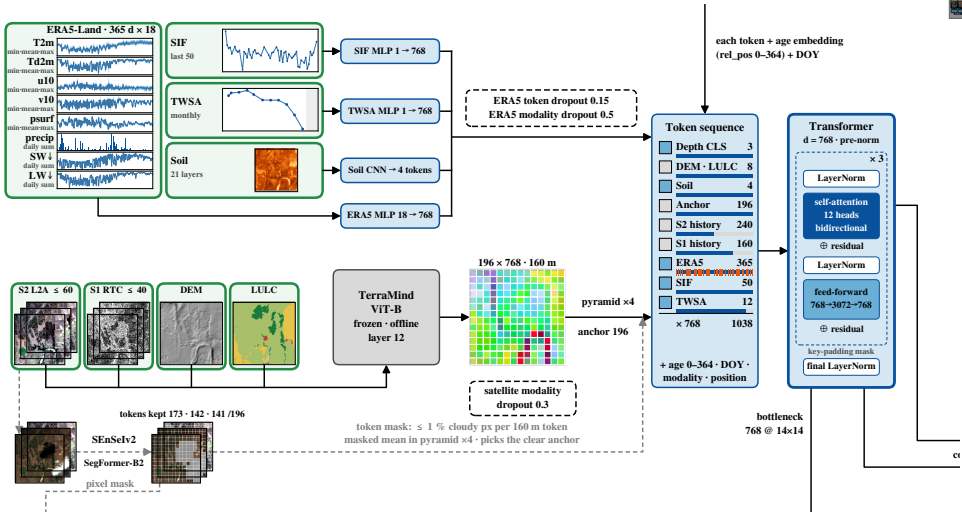
Method Inputs: what the model sees for day D



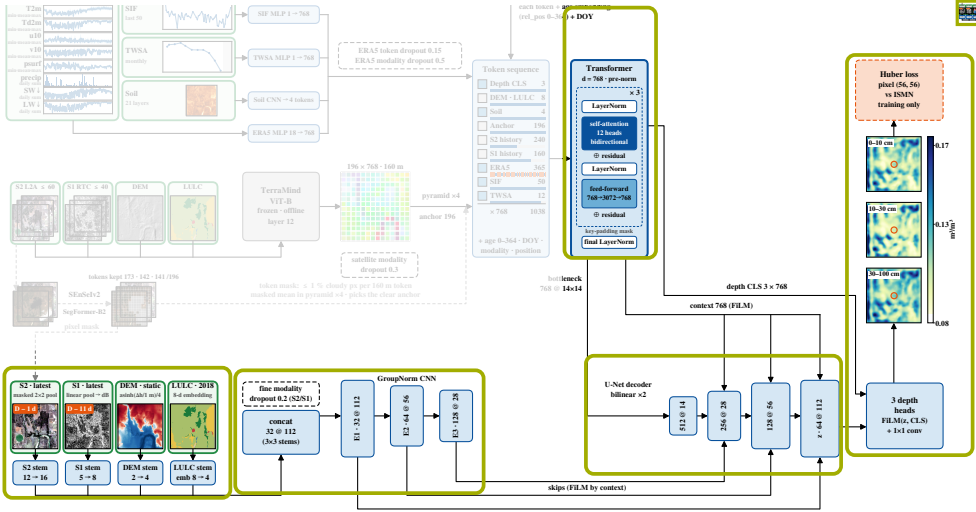


Method Coarse path: tokens → transformer

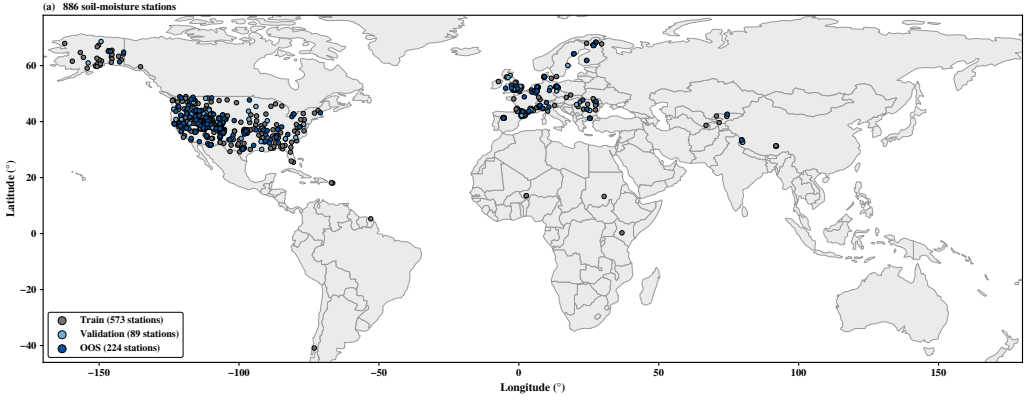




Method Fine path: 20 m CNN → U-Net decoder → depth heads



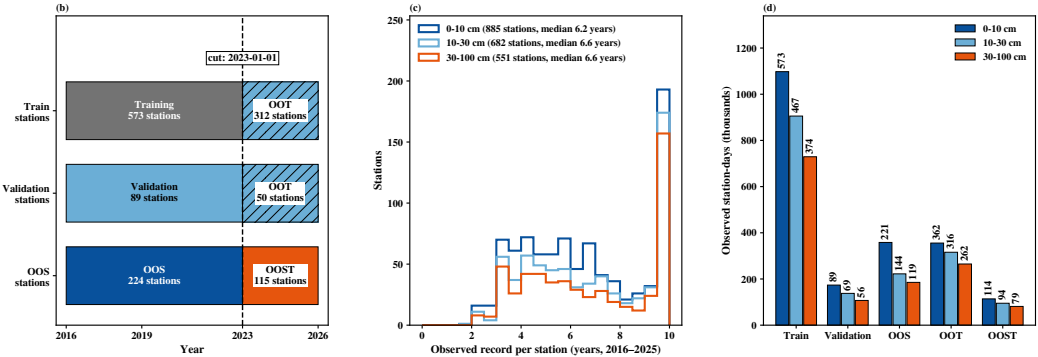
Data In-situ soil moisture: global coverage



886 soil-moisture stations (ISMN and flux sites): train 573, validation 89, out-of-sample (OOS) 224

Dense in North America and Europe; sparse in the tropics and the Southern Hemisphere

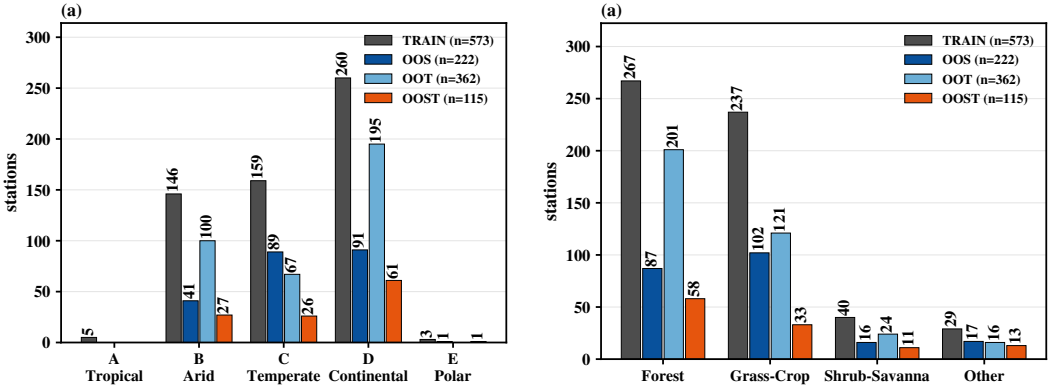
Data Depth coverage, record length and splits



(b) train / validation / OOS stations, time cut 2023-01-01; (c) observed record per station; (d) observed station-days

0–10 cm: 885 stations (median 6.2 yr); 10–30 cm: 682 (6.6 yr); 30–100 cm: 551 (6.6 yr)

Data Climate and land-cover coverage



Stations per Köppen climate zone (left) and land-cover class (right), by split

Mostly continental, temperate and arid; forest and grass–crop; almost no tropics or polar

Stations	990 in inventory; **886** with soil moisture (train 573 / val 89 / OOS 224)
Depths	0–10 cm: 885 10–30 cm: 682 30–100 cm: 551 stations
Period	2016–2025; out-of-time test from 2023-01-01
Soil-moisture observations	**5.14 M** observed daily values (2.10 M / 1.68 M / 1.37 M per depth)
Sentinel-2 L2A	272,281 scenes (994 sites); cloud $< 75\,\%$ per scene, cloudy tokens masked
Sentinel-1 RTC	496,835 scenes: 251,310 ascending + 245,525 descending
ERA5-Land	18 variables, 2.46 M station-days (990 stations)
Static / other	DEM, LULC, soil (OpenLandMap), SIF, TWSA

About 5.1 million daily soil-moisture observations at 886 stations, 0–100 cm